

Scalability & Future Plan

Date	2 July 2024
Team ID	XXXXXX
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Flood prediction relies on 5 manually entered input values rather than live sensor/satellite data	Limits real-time accuracy and depends on the user providing correct inputs	Medium
2	Weather alert scheduler (APScheduler) runs inside the same Flask process	Alerts stop firing if the web process restarts or the service scales down	High
3	Single web service instance (2 gunicorn workers, no horizontal scaling)	Limited capacity to handle high concurrent traffic during peak usage	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Render web service running gunicorn with 2 workers	Scale to multiple instances/workers behind a load balancer as usage grows
2	Data Storage	Managed PostgreSQL database on Render	Add connection pooling and read replicas as data volume and user base grow
3	Model Inference	Models loaded from disk and cached in memory per process	Move to a dedicated model-serving layer for shared caching and faster inference across instances
4	Background Alerts	APScheduler running hourly checks inside the Flask process	Move to an external job queue (e.g., Celery + Redis) for more reliable, scalable alert delivery

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integrate live weather/sensor data directly into the prediction pipeline	Q4 2026	Reduce dependency on manually entered input values
Phase 2	Move background alert checks to a dedicated worker/queue service	Q1 2027	Improve reliability of automated weather alerts
Phase 3	Add SMS-based alerts alongside email notifications	Q2 2027	Increase accessibility for users without regular email access
Phase 4	Multi-instance deployment with horizontal scaling	Q3 2027	Support a larger user base with better uptime and lower latency